

## Smart Mobile Inclusive Learning (SMILE): AI-driven multipurposed therapy for disability children - Indonesia - \$5935

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*EPICS in IEEE - Proposals*

### *IEEE Indonesian Section*

Ms Endah Suryawati Ningrum

M: 085330296329

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## FollowUp Form

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### *Project Participants*

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#### **Project Title, Country, Amount Requested\***

Please limit your Project Title to 100 characters.

Smart Mobile Inclusive Learning (SMILE): AI-driven multipurposed therapy for disability children - Indonesia - \$5935

#### **Number of total university students (undergraduate & graduate) directly involved in the project?\***

10

#### **Number of women university students (undergraduate & graduate) directly involved in the project?\***

5

#### **Number of men university students (undergraduate & graduate) directly involved in the project?\***

5

#### **How many of the undergraduate and graduate students are IEEE student members?\***

5

#### **Please list the names, emails and member-IDs (if known) here of all the students on the team.\***

Please use their personal emails versus school emails when possible.

Endah Suryawati Ningrum, endahpens@gmail.com,  
Gede Aditra Pradnyana, gede.aditra@gmail.com, 98630118  
Farah Zakiyah Rahmanti, fzrahmanti@gmail.com, 98664232

Fayruz Rahma, fayruzzrahma@gmail.com, 98664186  
Moch. Iskandar Riansyah, 98636069

## Date that the student survey was sent to student team members.\*

Please send a survey to all students involved in the project so that the committee can get their feedback on their experience. The link to the survey is here: <https://forms.gle/kxHpAkLZYJD15tDE9>.

Confirm when the survey was sent out by adding the date the survey was sent below.

04/24/2024

## Number of total high school/intermediate/secondary school students directly involved in the project?\*

20

## Number of women/girl high school/intermediate/secondary school students directly involved?\*

10

## Number of men/boy high school/intermediate/secondary school students directly involved?\*

10

## Which IEEE chapters/sections contributed volunteers to this project\*

IEEE Indonesia Section

## Number of IEEE Young Professionals volunteers\*

0

## Number of IEEE Life Member volunteers\*

1

## Number of total IEEE professional (non-student) member volunteers\*

Including young professionals and life members

1

## Number of woman IEEE professional member volunteers\*

0

## *Industry Involvement in the Project*

### Number of individuals from industry involved as partners or mentors\*

5

## *Project Impact*

### Did this project culminate in a prototype that was tested?\*

N/A is a potential outcome for certain types of projects, such as an education outreach projects that focuses on an experience or event

Yes

If no, why not?

(if yes or n/a) Was this project deployed or implemented in the community?

Yes

If no, why not?

### How many people will be impacted in the first year of the project deployment?\*

Only answer if you answered n/a to the first impact question, or Yes to the prior question

20

## How likely is it that your EPICS in IEEE community solution will still be active in three years?\*

[Answer choices: Very likely, Somewhat likely, Not very likely, Not at all likely, Project was not designed for long-term deployment]

Somewhat likely

## How many people will be impacted in the first 3 years of the project deployment?\*

Only answer if you put "Project was not designed for long-term deployment" for the previous question

50

## Please provide additional details about how you impacted the community.\*

Did you use any formal methods to assess this? Are there any firsthand accounts from those impacted that you can share with EPICS in IEEE? How has the community reacted to the project?

We have created Smart Mobile Inclusive Learning (SMILE), an AI-driven multipurpose therapy for disabled children, to assist children with disabilities. Our development revolves around making hand therapy a routine task and an enjoyable experience. By incorporating fun, interactive elements into the therapy process, we've captured the children's interest and maintained their engagement over time. This approach has significantly improved their willingness to participate in regular therapy sessions, which is crucial for their progress.

To assess the impact of the system, we observed the children when they tried the game therapy and interviewed the therapists. The children's and therapists' feedback has been positive, with many emphasizing how our system has not only aided in the physical development of the children but has also brought joy and lightness into what was once considered a strenuous task. Our system has demonstrated that treatment can be effective and fun simultaneously, setting a precedent for future innovations in therapeutic practices for children with disabilities.

Our team's doctors, therapists, and professionals have discussed further research opportunities to make physiotherapy better and more fun. This project has not only helped children with disabilities. It has also enriched the community's approach to disability and therapy, fostering a more inclusive, understanding, and supportive environment.

## Please add the community partner contact for your project\*

EPICS in IEEE would like to send a survey to your community partner. Please provide an email address if available.

We will copy the project lead on the email when the survey is sent. *(If you prefer to send the survey directly here is the link, please indicate that it was sent in the space below: <https://forms.gle/rsj1W8YD4SJ73H2r9>)*

We sent the survey link to the community partner

To be completed by the student lead of the project.

For following questions, likert style answers should include the following range: Significantly impacted, Moderately Impacted, Somewhat Impact, Minimal Impact, No impact.

For the following questions, rate how much impact the the following had on you.

- 1.) No impact
- 2.) Minimal impact
- 3.) Somewhat impacted
- 4.) Moderately impacted
- 5.) Significantly impacted

To what extent has the EPICS in IEEE project contributed to your development in the following areas:

## Project Management\*

Max Score: 5

4

## Problem Solving\*

Max Score: 5

4

## Engineering Design Process\*

Max Score: 5

4

## Increased understanding of your technical discipline\*

Max Score: 5

4

## Increased understanding of other technical disciplines\*

Max Score: 5

4

## Incorporating stakeholder and/or community feedback into the design process\*

Max Score: 5

4

To what extent has the EPICS in IEEE project contributed to your development in the following areas

## Leadership\*

Max Score: 5

4

## Teamwork\*

Max Score: 5

4

## Written Communication\*

Max Score: 5

4

## Verbal or Oral Communication\*

Max Score: 5

4

To what extent has the EPICS in IEEE project contributed to your development in the following areas:

# EPICS<sup>IN</sup>IEEE

## Understanding the needs of the end-user\*

Max Score: 5

4

## Social-Impact\*

Max Score: 5

4

## Empathy\*

Max Score: 5

4

## Personal Development

Please provide a brief example if there is any particular experience that significantly contributed to your learning or personal development during this project.

If you have any additional feedback that you would like to submit to EPICS in IEEE about your personal experience on this EPICS in IEEE project, please contact EPICS in IEEE at [epicsinieee@ieee.org](mailto:epicsinieee@ieee.org)

## Project Summary and Budget

### Please include links to or uploads of documentation for your prototype.\*

Our committee members are excited to see what you have built. Please share links or upload a document showcasing the prototype. This can include technical schematics, pictures of the prototype being tested, video footage, or presentations given about the project.

English Ver - SMART MOBILE INCLUSIVE LEARNING.docx

YouTube video: <https://youtu.be/5RnoNKx5GTo>

SMILE user guide is attached.

## Summary\*

Please provide a brief project summary of what was accomplished during this project and how the various students, volunteers, and community members interacted. You can include a summary of any major changes in project scope, engagement, or participation as well



This project, aimed at developing an AI-driven multipurpose therapy system for children with disabilities in YPAC Surabaya, successfully achieved its primary objectives. We created the system to provide cognitive and psychomotor therapy for children with mental retardation using advanced technologies like computer vision.

During the project, diverse participants, including university students, IEEE volunteers, and community members, worked together to develop, test, and implement the therapy system. All the participants engaged in the project. We collaborated with YPAC Surabaya for data collection, system design, and testing. There were no major changes in the project scope. However, there were minor changes, such as the OS used in the system, the game story, etc.

There are suggestions to create another, more dynamic version of the game because children might get bored after playing it for about 1-2 weeks. This advice is sound and constructive. We provide learning videos as rewards for children who have completed the game. We also offer a system for therapists and parents to monitor the children's progress and therapy history. The community has accepted the system well.

Overall, the project positively impacted approximately 20 children, providing them with innovative and accessible therapy options. The system developed is expected to improve the quality of therapy available to children with disabilities, enabling YPAC Surabaya to offer more engaging and effective therapy services. The success of the project also sets the stage for potential future expansions to other YPAC branches in different cities.

## Please upload a copy of your final budget\*

EPICS SMILE Expenditure- eng.pdf

## Additional Funding?\*

Did you have any other funding (financial or equipment) from universities, companies, grants, or private-donations outside of the EPICS in IEEE grant

No

If yes, please list these sources and approximate values

## Publicity

### Has this project been advertised or publicized in any articles or stories?\*

Sample publicity may include university-hosted web articles, social media shout-out, local news videos, printed magazine article, etc.

Yes

## If yes, please share the links or upload a copy of the materials with us!

See the upload button below.

1. ITS News - EN: <https://www.its.ac.id/news/en/2024/06/19/its-develops-physical-therapy-tool-receives-epics-ieee-grant/>
2. ITS News - ID: <https://www.its.ac.id/news/2024/06/19/kembangkan-alat-terapi-fisik-tim-its-terima-hibah-epics-ieee/>
3. ITS News - IG: <https://www.instagram.com/p/C8ZUJyQSNAC>
4. Electics News - EN: <https://www.its.ac.id/bringing-smiles-its-students-ai-innovation-for-therapy-and-learning-for-children-with-special-needs/>
5. Electics News - ID: <https://www.its.ac.id/id/menghadirkan-senyumai-inovasi-ai-mahasiswa-its-untuk-terapi-dan-pembelajaran-anak-berkebutuhan-khusus/>

## Did this project receive any awards?

If yes, please share any information or publicity links with us!

No

## Team Member's Stories

Would you like to recommend or nominate any team members or community partners who could be interviewed to tell their story about your project? If yes, please provide their contact information here

## Do you have any general comments or other feedback on this final reporting form?

## File Attachment Summary

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### *Applicant File Uploads*

- English Ver - SMART MOBILE INCLUSIVE LEARNING.docx
- EPICS SMILE Expenditure- eng.pdf

# SMART MOBILE INCLUSIVE LEARNING (SMILE): AI-DRIVEN MULTIPURPOSED THERAPY FOR DISABLED CHILDREN

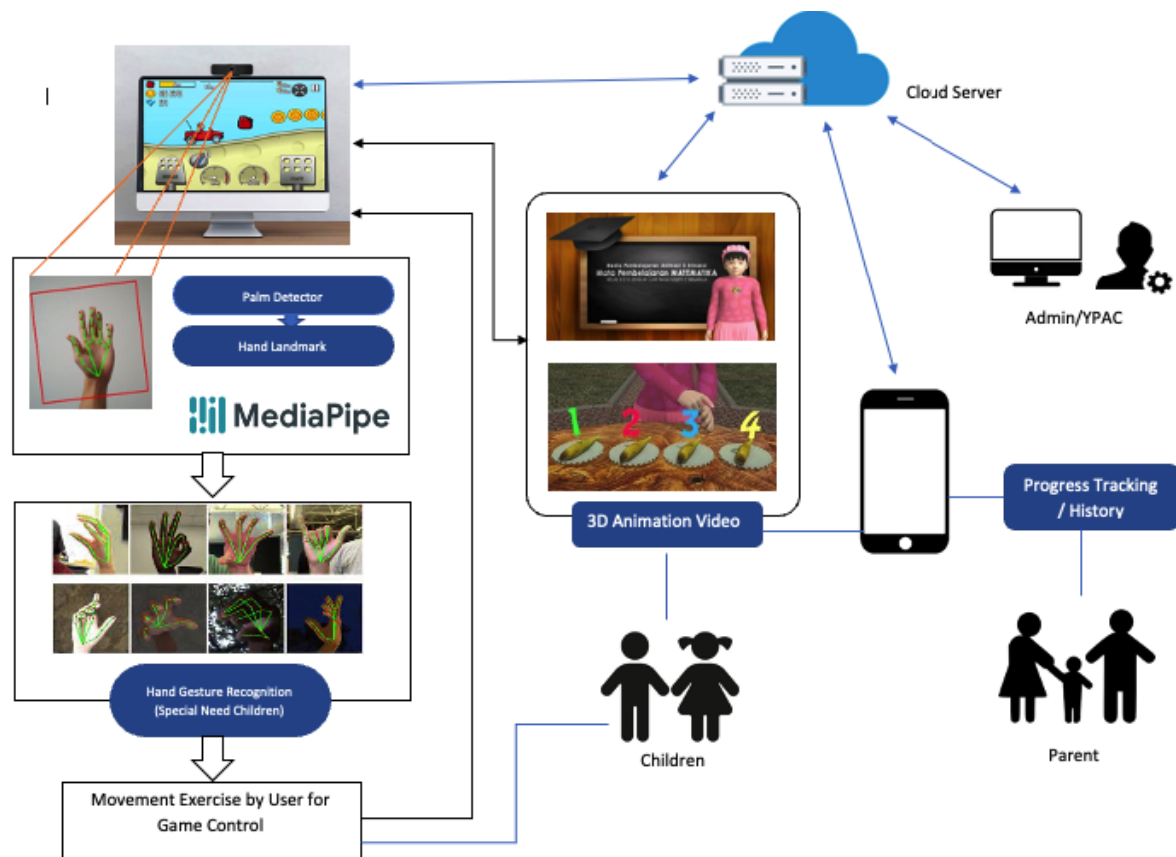


Moch. Iskandar Riansyah  
Gede Aditra Pradnyana  
Endah Suryawati Ningrum  
Fayruz Rahma  
Farah Zakiyah Rahmanti

2024

SMILE App Description

SMILE is a therapy tool for children with psychomotor and cognitive disabilities. Equipped with a camera, mini PC, and monitor screen, SMILE uses machine learning methods for pose detection and adaptive 3D animation-based audio-visual media according to the child's therapy needs. This device combines aspects of treatment and play to make the therapy process more interesting and effective.



**Figure 1. SMILE System Architecture**

Figure 1 illustrates the SMILE architecture developed for children with special needs using artificial intelligence technology. Here is the explanation of each component in the figure:

1. **Monitor with an Integrated Camera:**
  - o Children interact with games or objects displayed on the monitor. A camera integrated into the monitor captures the child's movements as they interact.
2. **MediaPipe:**
  - o **Palm Detector and Hand Landmark:** MediaPipe is used to detect a child's hand pose with high accuracy. It includes palm detection and hand landmark marking.
  - o **Hand Gesture Recognition:** This technology recognizes the hand movements of children with special needs to control games. These recognized movements control the game displayed on the monitor screen.
3. **Therapy movement:**

- Children perform movement exercises as part of game control. These movements assist in psychomotor therapy designed to improve finger and hand coordination.
- 4. **3D Animation Video:**
  - 3D animation media is used as part of cognitive therapy. This animation is made with Blender and is tailored to the needs of children. This animation includes basic math learning, such as addition and subtraction.
  - **Integration with Psychomotor Therapy:** 3D animation videos are also integrated with psychomotor therapy tools. For example, children can receive rewards in the form of animation when they complete certain movements in the game.
- 5. **Cloud Server:**
  - SMILE information system and data are stored on cloud servers for easy access and mobility. These cloud servers enable efficient data storage and management.
- 6. **Admin/YPAC:**
  - YPAC administrators or staff can access the system via computer to monitor and manage the therapy process.
- 7. **Parents:**
  - Parents can track their child's progress and therapy history through a web application on their smartphone.
- 8. **Children:**
  - Children interact or do physiotherapy directly with the SMILE game through movement exercises recorded by the camera and through the 3D animated video content provided.

With SMILE, the therapy process becomes more interesting and effective for children with special needs, especially those with mental retardation disorders. This system combines psychomotor and cognitive aspects with a gamification approach to increase engagement and therapy results. The concept of gamification is also applied to make the therapy process more interesting for children.

## Program User Manual:

1. Run the program file (SMILE\_Game.py) using the Visual Studio Code IDE . Click **Run** on Visual Studio Code until the initial game display appears, as in Figure 2 below.

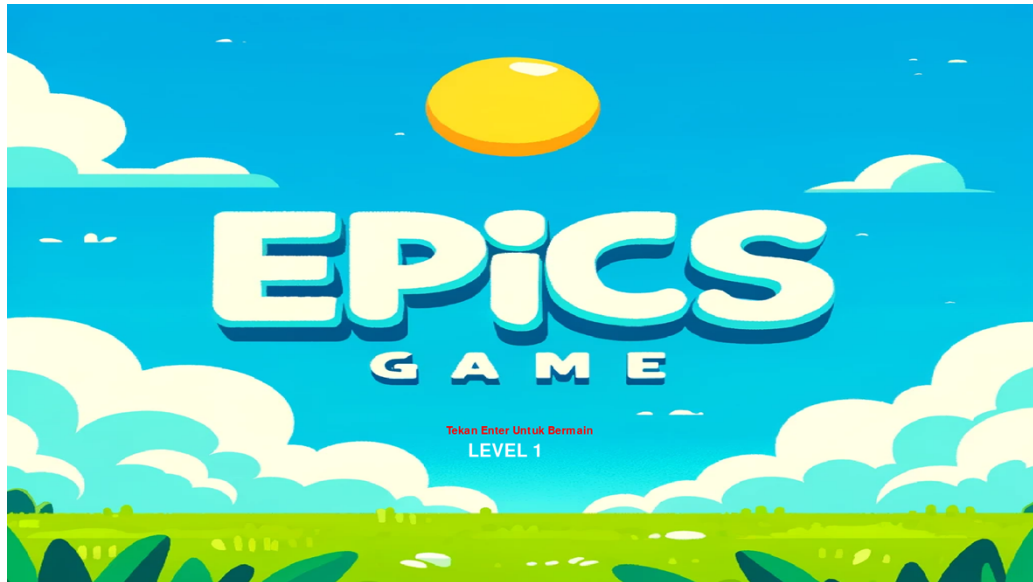


Figure 2. SMILE Game Start Screen

2. After the display appears on the computer screen, as in Figure 2, you will see the words "Tekan Enter Untuk Bermain" ("Press Enter to Play") and "LEVEL 1". As in Figure 3 below.

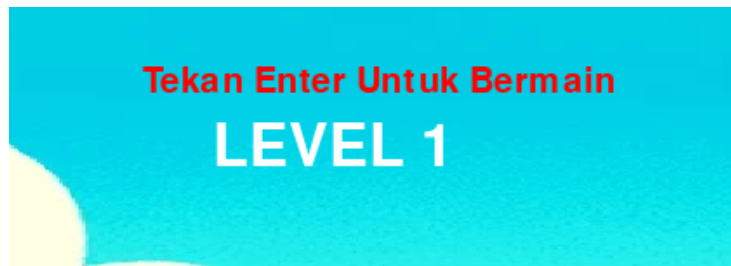
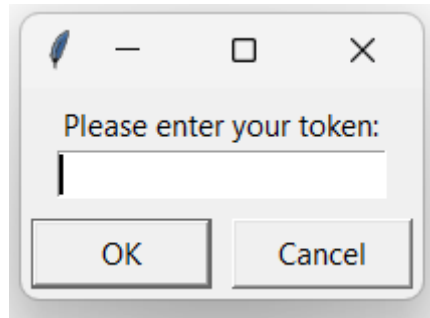


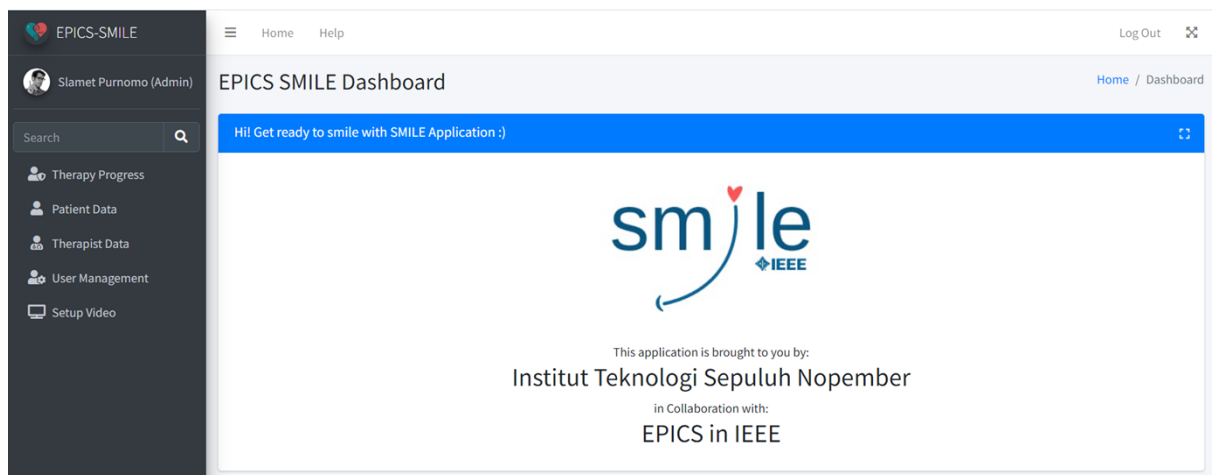
Figure 3. Game LEVEL Selection Menu

3. In this game, if you want to choose a different level, press the up or down arrow keys on the keyboard to lower and raise the level so that the words "LEVEL 1" will change. There are three levels: LEVEL 1, LEVEL 2, and LEVEL 3.
4. After selecting the desired level for therapy, press Enter on your computer keyboard, and a dialog box will appear (as shown in Figure 4 below) to enter the token of the patient being treated.



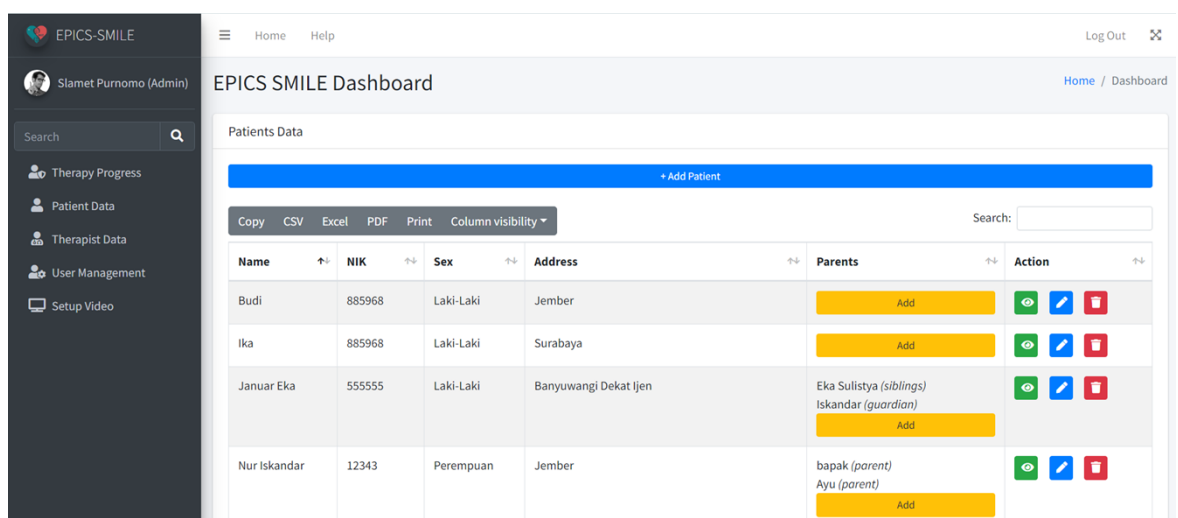
**Figure 4.** The dialog box for entering the token

5. You can obtain the token from the database using the SMILE web application (<https://epics-smile.id/>). Please log in using your username and password.
6. You will see a web dashboard that looks like the following (see Figure 5).



**Figure 5.** SMILE Web Application Dashboard View

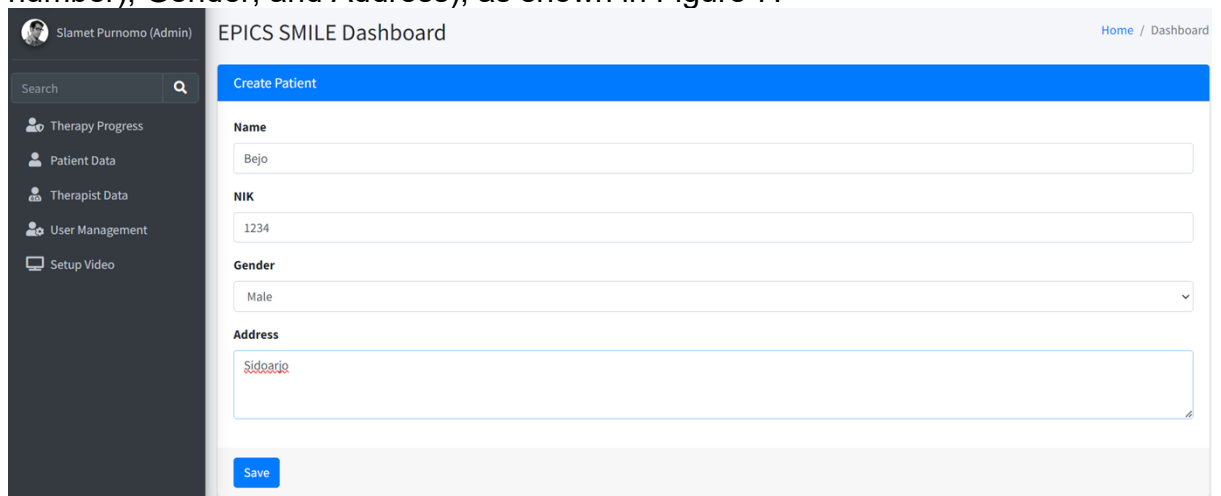
7. On the left side, there are several menu options. Click the "Patient Data" and "Add Patient" menus to add patient data to the database.



**Figure 6.** "Add Patient" Menu Display



8. After that, a display will appear to enter patient data (Name, NIK (national ID number), Gender, and Address), as shown in Figure 7.



EPICS SMILE Dashboard

Home / Dashboard

Create Patient

Name: Bejo

NIK: 1234

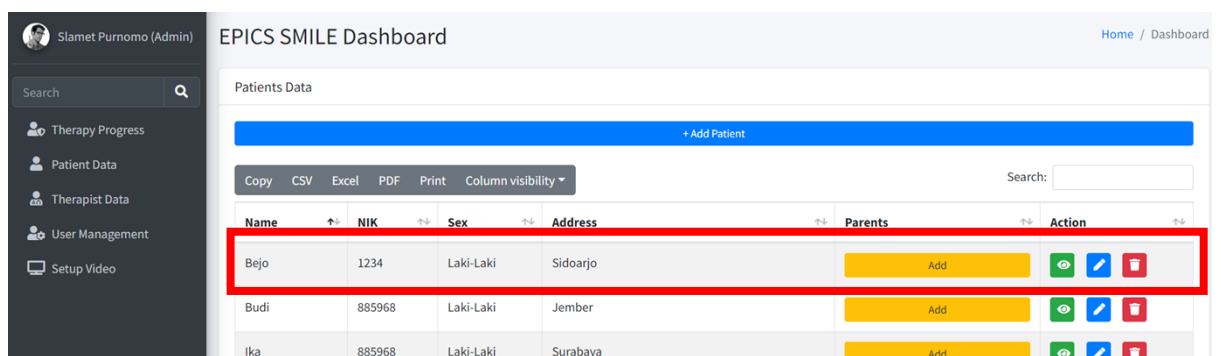
Gender: Male

Address: Sidoarjo

Save

Figure 7. View for entering patient data

9. Next, click "Save" to save the data. The system will save the data you just entered into the web database.



EPICS SMILE Dashboard

Home / Dashboard

Patients Data

+ Add Patient

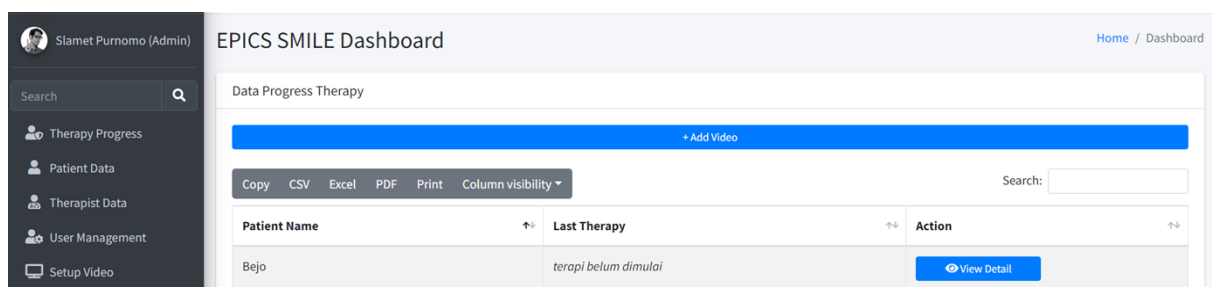
Copy CSV Excel PDF Print Column visibility

Search:

| Name | NIK    | Sex       | Address  | Parents | Action           |
|------|--------|-----------|----------|---------|------------------|
| Bejo | 1234   | Laki-Laki | Sidoarjo | Add     | View Edit Delete |
| Budi | 885968 | Laki-Laki | Jember   | Add     | View Edit Delete |
| Ika  | 885968 | Laki-Laki | Surabaya | Add     | View Edit Delete |

Figure 8. New patient data appears in the web database.

10. Then, return to the selection menu on the left side of the dashboard and select the "Therapy Progress" menu. The display will show that the patient has yet to undergo SMILE therapy, as shown in Figure 9.



EPICS SMILE Dashboard

Home / Dashboard

Data Progress Therapy

+ Add Video

Copy CSV Excel PDF Print Column visibility

Search:

| Patient Name | Last Therapy         | Action      |
|--------------|----------------------|-------------|
| Bejo         | terapi belum dimulai | View Detail |

Figure 9. "Therapy Progress" Menu Display

11. Next, select the Action menu "View Detail" and add therapy by choosing the "Add Therapy" menu.

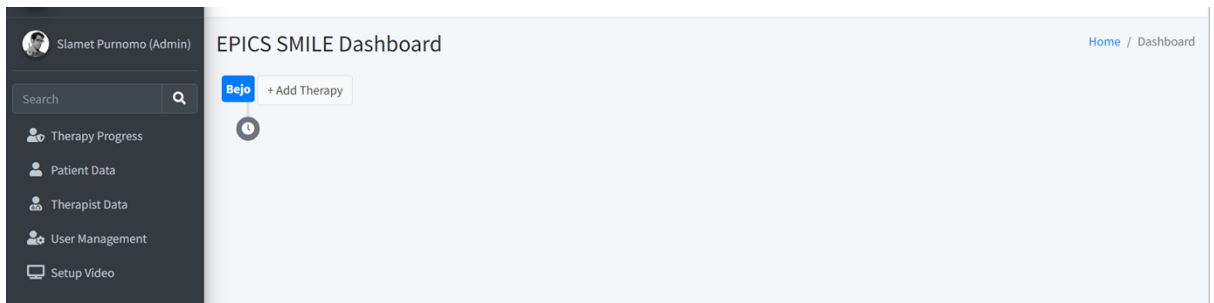


Figure 10. View Detail of Patient Data

12. Enter the therapy data as shown in Figure 11. Then, click "Save" to save the therapy data.

Figure 11. Display for entering patient therapy data

13. After you save it, token data will appear from patient data registered to go through the therapy process using the SMILE game.

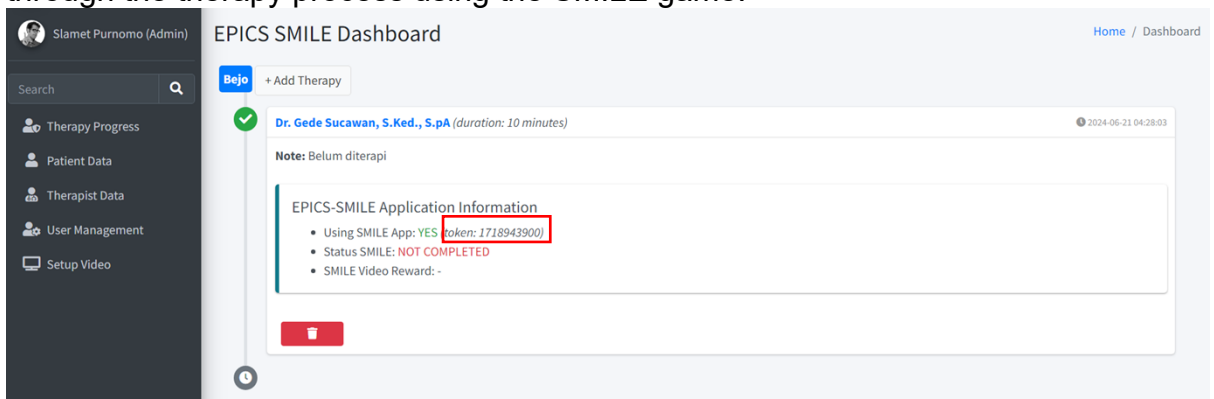
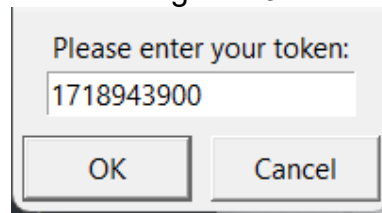


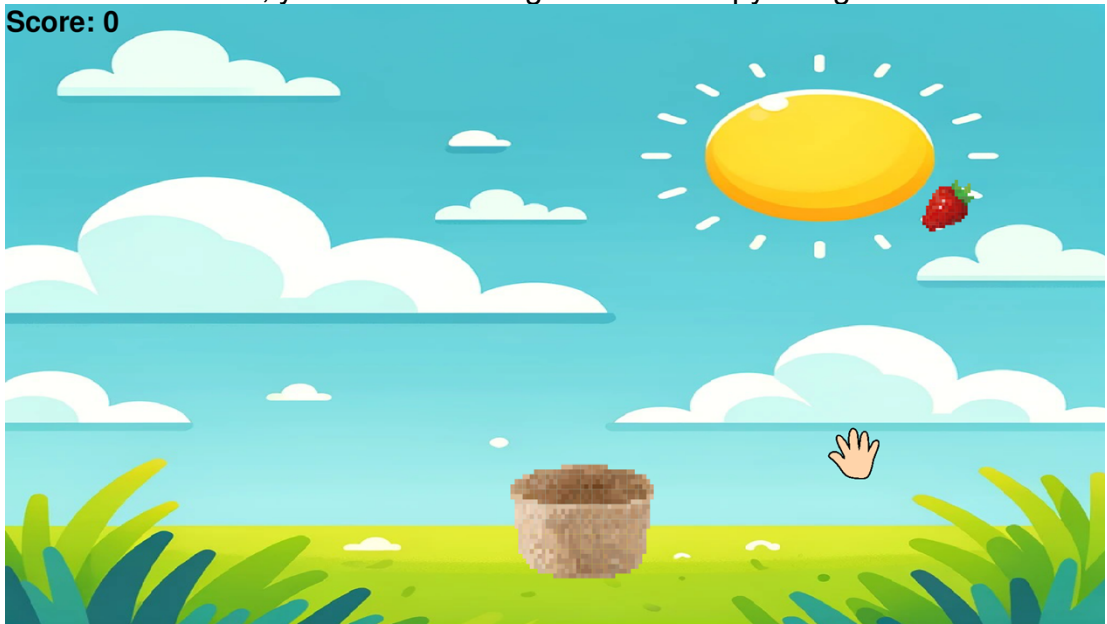
Figure 12. Token for each registered patient

14. Next, enter the token into the dialog box on your computer desktop (on the SMILE game screen), as shown in Figure 13.



**Figure 13.** Enter the token into the dialog box before playing the game.

15. Select "OK". Then, you will enter the game for therapy using hand movements.  
**Score: 0**



**Figure 14.** SMILE therapy game uses hand movements



**Figure 15.** Therapy child is playing the SMILE game

16. After the patient finishes the challenges of putting objects into the basket, the game will send the therapy data to the web. The therapy status will change to "COMPLETE." Then, the next therapy guide will appear, which is to watch an educational video as the next therapy for the cognitive aspect.

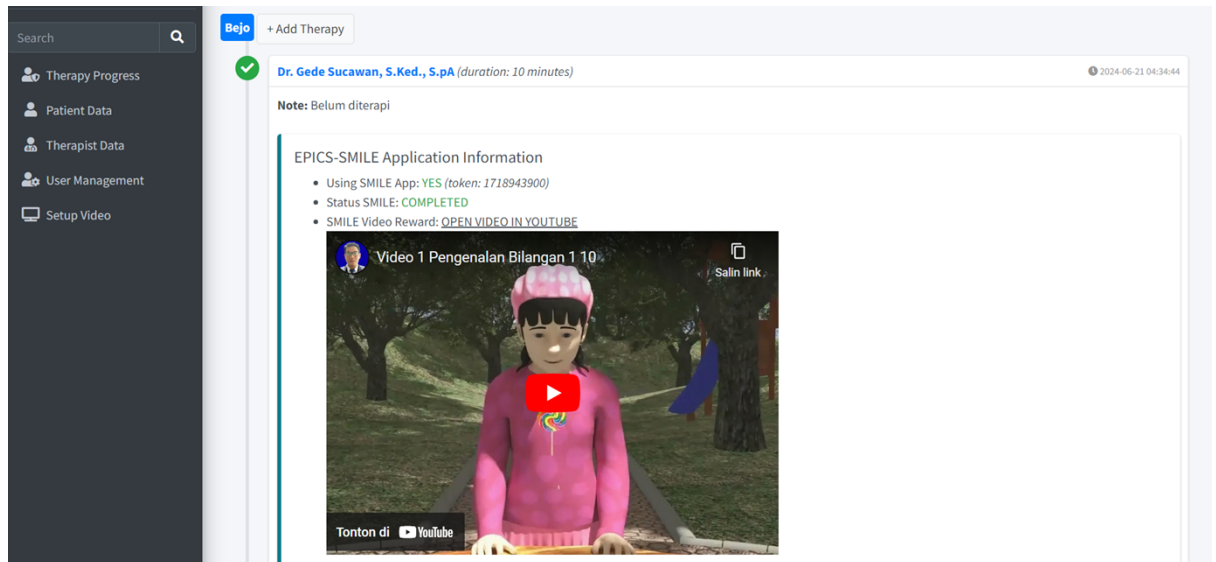


Figure 15. The physiotherapy process with games is complete, and the final stage is watching an educational video.

|    |              | SMILE Project Expenditure                                            |          |                                 |              |
|----|--------------|----------------------------------------------------------------------|----------|---------------------------------|--------------|
|    |              |                                                                      |          |                                 |              |
| No | Date         | Expense Details                                                      | Quantity | Price                           | Total        |
| 1  |              | Stationery                                                           | 4        | Rp1,700,000                     | Rp6,800,000  |
| 2  |              | Event Consumption (Interview): Hokben (Rice Box)                     | 1        | Rp559,000                       | Rp559,000    |
| 3  |              | Transportation (Interview)                                           | 3        | Rp200,000                       | Rp600,000    |
| 4  |              | Event Consumption (Data Collection): Hokben (Rice Box)               | 1        | Rp559,000                       | Rp559,000    |
| 5  |              | Transportation (Data Collection)                                     | 3        | Rp200,000                       | Rp600,000    |
| 6  |              | FGD SMILE Project Preparation                                        | 1        | Rp3,000,000                     | Rp3,000,000  |
| 7  |              | Literature References                                                | 6        | Rp150,000                       | Rp900,000    |
| 8  | Nov 21, 2023 | Hosting and Domain during 1 year                                     | 1        | Rp733,602                       | Rp733,602    |
| 9  |              | Supporting Electrical Equipment (Solder, Socket, Drill)              | 1        | Rp4,500,000                     | Rp4,500,000  |
| 10 |              | Cable                                                                | 2        | Rp250,000                       | Rp500,000    |
| 11 | Dec 21, 2023 | Bracket TV Standing Monitor                                          | 3        | Rp434,100                       | Rp1,302,300  |
| 12 | Dec 21, 2023 | GEEKOM Mini PC                                                       | 3        | Rp5,174,800                     | Rp15,524,400 |
| 13 | Dec 21, 2023 | Webcam Logitech C992 PRO STREAM FUL HD                               | 3        | Rp1,478,600                     | Rp4,435,800  |
| 14 | Dec 21, 2023 | LED Monitor LG                                                       | 3        | Rp1,966,800                     | Rp5,900,400  |
| 15 |              | Keyboard Logitech K270                                               | 3        | Rp239,200                       | Rp717,600    |
| 16 |              | Mouse Wireless Logitech M170                                         | 3        | Rp122,500                       | Rp367,500    |
| 17 |              | OPPO A3x 4GB/64GB                                                    | 3        | Rp1,599,000                     | Rp4,797,000  |
| 18 | Jan 30, 2024 | Event Consumption (Updating Progress to YPAC): Hokben (Rice Box)     | 1        | Rp559,000                       | Rp559,000    |
| 19 |              | Transportation (Updating Progress to YPAC)                           | 3        | Rp200,000                       | Rp600,000    |
| 20 | May 14, 2024 | Event Consumption (Updating Progress to YPAC): Kampoeng roti (Snack) | 1        | Rp189,000                       | Rp189,000    |
| 21 |              | Transportation (Updating Progress to YPAC)                           | 3        | Rp200,000                       | Rp600,000    |
| 22 | May 16, 2024 | Xbanner                                                              | 1        | Rp60,000                        | Rp60,000     |
| 23 | May 29, 2024 | Deliver Xbanner: Gosend to SMILE Project Exhibition                  | 1        | Rp23,500                        | Rp23,500     |
| 24 |              | Transportation SMILE Exhibition                                      | 3        | Rp200,000                       | Rp600,000    |
| 25 | May 29, 2024 | Handover Event Consumption: Hokben (Rice Box)                        | 1        | Rp559,000                       | Rp559,000    |
| 26 |              | Transportation Handover Event Consumption                            | 3        | Rp200,000                       | Rp600,000    |
| 27 |              | Printing User Guide                                                  | 30       | Rp50,000                        | Rp1,500,000  |
| 28 |              | Rent Space for Handover Event Consumption                            | 1        | Rp1,950,000                     | Rp1,950,000  |
| 29 | Jun 13, 2024 | EPIC T-Shirt (8 people)                                              | 1        | Rp799,000                       | Rp799,000    |
| 30 | Jul 30, 2024 | Copyright (HKI) Payment                                              | 1        | Rp600,000                       | Rp600,000    |
| 31 |              | Mass Media Article Costs                                             | 4        | Rp700,000                       | Rp2,800,000  |
| 32 |              | Community Service Journal Publication                                | 1        | Rp1,800,000                     | Rp1,800,000  |
| 33 |              | Logo, Poster, Xbanner Design                                         | 3        | Rp165,000                       | Rp495,000    |
| 34 |              | Learning Video                                                       | 10       | Rp1,600,000                     | Rp16,000,000 |
| 35 |              | Video Documentation                                                  | 1        | Rp1,600,000                     | Rp1,600,000  |
|    |              | Total                                                                |          |                                 | Rp83,131,102 |
|    |              |                                                                      |          | (in Indonesian Rupiah currency) |              |